
Small Language Models are the Future of Agentic AI

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Abstract

Large language models (LLMs) are often praised for exhibiting near-human performance on a wide range of tasks and valued for their ability to hold a general conversation. The rise of agentic AI systems is, however, ushering in a mass of applications in which language models perform a small number of specialized tasks repetitively and with little variation.

Here we lay out the position that small language models (SLMs) are *sufficiently powerful, inherently more suitable, and necessarily more economical for many invocations in agentic systems, and are therefore the future of agentic AI*. Our argumentation is grounded in the current level of capabilities exhibited by SLMs, the common architectures of agentic systems, and the economy of LM deployment. We further argue that in situations where general-purpose conversational abilities are essential, heterogeneous agentic systems (i.e., agents invoking multiple different models) are the natural choice. We discuss the potential barriers for the adoption of SLMs in agentic systems and outline a general LLM-to-SLM agent conversion algorithm.

Our position¹, formulated as a value statement, highlights the significance of the operational and economic impact even a partial shift from LLMs to SLMs is to have on the AI agent industry. We aim to stimulate the discussion on the effective use of AI resources and hope to advance the efforts to lower the costs of AI of the present day. Calling for both contributions to and critique of our position, we commit to publishing all such correspondence at research.nvidia.com/labs/lpr/slm-agents.

1 Introduction

The deployment of agentic artificial intelligence is on a meteoric rise. Recent surveys show that more than a half of large IT enterprises are actively using AI agents, with 21% having adopted just within the last year [14]. Aside from the users, markets also see substantial economic value in AI agents: As of late 2024, the agentic AI sector had seen more than USD 2bn in startup funding, was valued at USD 5.2bn, and was expected to grow to nearly USD 200bn by 2034 [46, 51]. Put plainly, there is a growing expectation that AI agents will play a substantial role in the modern economy.

The core components powering most modern AI agents are (very) large language models [52, 48]. It is the LLMs that provide the foundational intelligence that enables agents to make strategic decisions about when and how to use available tools, control the flow of operations needed to complete tasks, and, if necessary, to break down complex tasks into manageable subtasks and to perform reasoning for action planning and problem-solving [52, 17]. A typical AI agent then simply communicates with

¹The views and positions expressed in this paper are those of the authors and do not necessarily reflect the views or positions of any entities they represent.

小型语言模型是智能体人工智能的未来

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Abstract

大型语言模型（LLMs）因其在广泛任务上展现出接近人类的性能而备受赞誉，并因其能够进行通用对话而受到重视。然而，智能体式人工智能系统的兴起，正在催生大量应用，其中语言模型重复执行少量专门化任务且变化很小。

在此，我们提出以下观点：小型语言模型（SLMs）是 *sufficiently powerful, inherently more suitable, and necessarily more economical for many invocations in agentic systems, and are therefore the future of agentic AI*。我们的论证基于当前 SLMs 所展现的能力水平、智能体系统的常见架构以及语言模型部署的经济性。我们进一步认为，在通用对话能力至关重要的情况下，异构智能体系统（即调用多个不同模型的智能体）是自然的选择。我们讨论了在智能体系统中采用 SLMs 的潜在障碍，并概述了一个通用的 LLM 到 SLM 智能体转换算法。

我们的立场¹，表述为一项价值主张，强调了即使是从 LLMs 部分转向 SLMs，也将对 AI 智能体行业产生的运营和经济影响的重要性。我们的目标是激发关于有效利用 AI 资源的讨论，并希望推动当前降低 AI 成本的努力。我们呼吁对我们的立场提出贡献和批评，并承诺将所有此类通信发表在 research.nvidia.com/labs/lpr/slm-agents。

1 Introduction

智能体人工智能的部署正呈爆发式增长。最近的调查显示，超过一半的大型 IT 企业正在积极使用 AI 智能体，其中 21% 是在过去一年内才采用的 [14]。除了用户之外，市场也看到了 AI 智能体巨大的经济价值：截至 2024 年底，智能体 AI 领域已获得超过 20 亿美元的初创企业融资，估值达到 52 亿美元，预计到 2034 年将增长至近 2000 亿美元 [46, 51]。简而言之，人们越来越期望 AI 智能体将在现代经济中扮演重要角色。

为大多数现代 AI 智能体提供核心动力的组件是（超）大规模语言模型 [52, 48]。正是 LLMs 提供了基础智能，使智能体能够就何时以及如何使用可用工具做出战略决策，控制完成任务所需的操作流程，并在必要时将复杂任务分解为可管理的子任务，以及为行动规划和问题解

¹本文所表达的观点和立场仅代表作者本人，不一定反映其所代表的任何实体的观点或立场。

a chosen LLM API endpoint by making requests to centralized cloud infrastructure that hosts these models [52].

LLM API endpoints are specifically designed to serve a large volume of diverse requests using one generalist LLM. This operational model is deeply ingrained in the industry. In fact, it forms the foundation of substantial capital investment in the hosting cloud infrastructure – estimated at USD 57bn [16]. It is anticipated that this operational model will remain the cornerstone of the industry and that the large initial investment will deliver returns comparable to traditional software and internet solutions within 3-4 years [57].

In this work, we recognize the dominance of the standard operational model but verbally challenge one of its aspects, namely the custom that the agents’ requests to access language intelligence are – in spite of their comparative simplicity – handled by singleton choices of generalist LLMs. We state (Section 2), argue (Section 3), and defend (Section 4) the position that the **small, rather than large, language models are the future of agentic AI**. We, however, recognize the business commitment and the now-legacy praxis that is the cause for the contrary state of the present (Section 5). In remedy, we provide an outline of a conversion algorithm for the migration of agentic applications from LLMs to SLMs (Section 6), and call for a wider discussion (Section 7). If needed to concretize our stance, we attach a set of short case studies estimating the potential extent of LLM-to-SLM replacement in selected popular open-source agents (Appendix B).

2 Position

2.1 Definitions

For the purpose of concretizing our position, we use the following working definitions:

WD1 A *SLM* is a LM that can fit onto a common consumer electronic device and perform inference with latency sufficiently low to be practical when serving the agentic requests of one user.

WD2 An *LLM* is a LM that is not a *SLM*.

We justify the wording of these definitions in Appendix A, but note that their choice has little bearing on the essence of our position. We note that as of 2025, we would be comfortable with considering most models below 10bn parameters in size to be SLMs.

We use the words *agent* and *agentic system* interchangeably, preferring the former when emphasizing the software with some agency as a whole (e.g., “as seen in popular coding agents”) and the latter when highlighting the systems aspect of the agentic application as a sum of its components (e.g., “not all LMs of an agentic system are replaceable by SLMs”). For brevity, we focus on LMs as the bedrock of agentic applications and do not explicitly consider vision-language models, although we note that our position and most of our arguments readily extend to vision-language models as well.

2.2 Statement

We contend that SLMs are

V1 principally *sufficiently powerful* to handle language modeling errands of agentic applications;

V2 inherently *more operationally suitable* for use in agentic systems than LLMs;

V3 necessarily *more economical* for the vast majority of LM uses in agentic systems than their general-purpose LLM counterparts by the virtue of their smaller size;

and that on the basis of views V1–V3 SLMs are the future of agentic AI.

The phrasing of our position is deliberate. In its statement, we wish to convey that the described future development is ultimately a necessary consequence of the differences between SLMs and LLMs if the natural priorities are followed. We do not make a recommendation or try to impose an obligation — we make a statement of what we see as a faithful reflection of the community’s values in this context.

决进行推理 [52, 17]。一个典型的 AI 智能体随后只需通过向托管这些模型的集中式云基础设施发出请求，即可与选定的 LLM API 端点进行通信 [52]。

LLM API 端点专门设计用于使用一个通用型 LLM 来服务大量多样化的请求。这种运营模式在行业中根深蒂固。事实上，它构成了对托管云基础设施进行巨额资本投资的基础——估计达 570 亿美元 [16]。预计这种运营模式将继续成为行业的基石，并且巨大的初始投资将在 3-4 年内带来与传统软件和互联网解决方案相当的回报 [57]。

在这项工作中，我们承认标准运营模式的主导地位，但口头挑战其一个方面，即尽管智能体访问语言智能的请求相对简单，但习惯上仍由单一选择的通用型 LLM 来处理。我们陈述 (Section 2)、论证 (Section 3) 并捍卫 (Section 4) 这一立场：**small, rather than large, language models are the future of agentic AI**。然而，我们认识到导致当前相反状态 (Section 5) 的商业承诺和已成为惯例的做法。作为补救措施，我们提供了一个用于将智能体应用从 LLM 迁移到 SLM 的转换算法大纲 (Section 6)，并呼吁进行更广泛的讨论 (Section 7)。如果需要具体说明我们的立场，我们附上一组简短的案例研究，估算了在选定的流行开源智能体中 LLM 到 SLM 替换的潜在程度 (Appendix B)。

2 Position

2.1 Definitions

为了具体阐明我们的立场，我们采用以下工作定义：

WD1 一个 *SLM* 是指能够部署在常见消费电子设备上，并以足够低的延迟执行推理，从而在服务单个用户的智能体请求时具有实用性的语言模型。

WD2 一个 *LLM* 是指不属于 *SLM* 的语言模型。

我们在 Appendix A 中论证了这些定义措辞的合理性，但需注意，其具体选择对我们立场的核心影响甚微。我们注意到，截至 2025 年，我们倾向于将大多数参数规模低于 100 亿的模型视为 SLM。

我们交替使用 *agent* 和 *agentic system* 这两个词，当强调具有某种自主性的软件整体时（例如，“正如在流行的编码智能体中所见”）倾向于前者，而当强调智能体应用作为其组件总和的系统层面时（例如，“并非智能体系统中的所有语言模型都能被 SLM 替代”）则倾向于后者。为简洁起见，我们聚焦于语言模型作为智能体应用的基石，并未明确考虑视觉语言模型，尽管我们注意到我们的立场和大部分论点同样适用于视觉语言模型。

2.2 Statement

We contend that SLMs are

V1 主要 *sufficiently powerful* 以处理智能体应用中的语言建模任务；

V2 本质上 *more operationally suitable* 于在智能体系统中使用，相较于 LLM；

V3 由于其较小的规模，必然 *more economical* 于智能体系统中绝大多数语言模型的使用，相较于其通用的 LLM 对应物；

并且基于视图 V1–V3，SLM 是智能体 AI 的未来。

2.3 Elaboration

We assert that the dominance of LLMs in the design of AI agents is both excessive and misaligned with the functional demands of most agentic use cases. While LLMs offer impressive generality and conversational fluency, the majority of agentic *subtasks* in deployed agentic systems are repetitive, scoped, and non-conversational—calling for models that are efficient, predictable, and inexpensive. In this context, SLMs not only suffice, but are often preferable. They offer several advantages: lower latency, reduced memory and computational requirements, and significantly lower operational costs, all while maintaining adequate task performance in constrained domains.

Our position stems from a pragmatic view of language model usage patterns within agentic architectures. These systems typically decompose complex goals into modular sub-tasks, each of which can be reliably handled by specialized or fine-tuned SLMs. We argue that insisting on LLMs for all such tasks reflects a misallocation of computational resources—one that is economically inefficient and environmentally unsustainable at scale.

Moreover, in cases where general reasoning or open-domain dialogue is essential, we advocate for heterogeneous agentic systems, where SLMs are used by default and LLMs are invoked selectively and sparingly. This modular composition — combining the precision and efficiency of SLMs with the generality of LLMs — enables the construction of agents that are both cost-effective and capable.

Ultimately, we observe that shifting the paradigm from LLM-centric to SLM-first architectures represents to many not only a technical refinement but also a Humean moral ought. As the AI community grapples with rising infrastructure costs and environmental concerns, adopting and normalizing the use of SLMs in agentic workflows can play a crucial role in promoting responsible and sustainable AI deployment.

3 Position Arguments

We support views **V1–V3** by the following non-exclusive arguments.

3.1 SLMs are already sufficiently powerful for use in agents

A1 SLMs are sufficiently powerful to take the place of LLMs in agentic systems. This argument stands in support of view **V1**.

Over the past few years, the capabilities of small language models have advanced significantly. Although the LM scaling laws remain observed, the scaling curve between model size and capabilities is becoming increasingly steeper, implying that the capabilities of newer small language models are much closer to those of previous large language models. Indeed, recent advances show that well-designed small language models can meet or exceed the task performance previously attributed only to much larger models.

Extensive comparisons with large models have been conducted in the individual works cited below, but not all capabilities assessed by benchmarks are essential to their deployment in the agentic context. Here we highlight their aptitude for commonsense reasoning (an indicator of basic understanding), tool calling and code generation (both indicators of the ability to correctly communicate across the model→tool/code interface; see Figure 1; [77, 78]), and instruction following (ability to correctly respond back across the code←model interface; [83]). In each case, we also quote the efficiency increase if stated by the authors.

- **Microsoft Phi series.** Phi-2 (2.7bn) achieves commonsense reasoning scores and code generation scores on par with 30bn models while running $\sim 15\times$ faster [37]. Phi-3 small (7bn) [3] achieves language understanding and commonsense reasoning on par with and code generation scores running up to 70bn models of the same generation.
- **NVIDIA Nemotron-H family.** The 2/4.8/9bn hybrid Mamba-Transformer models achieve instruction following and code-generation accuracy comparable to dense 30bn LLMs of the same generation at an order-of-magnitude fraction of the inference FLOPs [9].
- **Huggingface SmoLLM2 series.** SmoLLM2 family of compact language models with sizes ranging from 125mn to 1.7bn parameters [6] each run up in their language understanding,

我们立场的措辞是经过深思熟虑的。在陈述中，我们希望传达的是，如果遵循自然的优先级，所描述的未来发展最终是 SLM 与 LLM 之间差异的必然结果。我们并非提出建议或试图强加义务——我们只是陈述我们认为在此背景下忠实反映社区价值观的观点。

2.3 Elaboration

我们断言，大语言模型在人工智能智能体设计中的主导地位既是过度的，也与大多数智能体应用场景的功能需求不匹配。虽然大语言模型提供了令人印象深刻的通用性和对话流畅性，但已部署智能体系统中的大多数智能体 *sub* 任务是重复的、范围明确且非对话式的——这需要高效、可预测且成本低廉的模型。在此背景下，小语言模型不仅足够胜任，而且通常是更优选择。它们提供了若干优势：更低的延迟、更少的内存和计算需求，以及显著降低的运营成本，同时在受限领域内保持足够的任务性能。

我们的立场源于对智能体架构中语言模型使用模式的务实看法。这些系统通常将复杂目标分解为模块化的子任务，每个子任务都可以由专门的或经过微调的小语言模型可靠地处理。我们认为，坚持为所有此类任务使用大语言模型反映了计算资源的错配——这在经济上是低效的，并且在规模上对环境是不可持续的。

此外，在通用推理或开放域对话至关重要的情况下，我们主张采用异构的智能体系统，其中默认使用小语言模型，而大语言模型则被选择性地、谨慎地调用。这种模块化组合——将小语言模型的精确性和效率与大语言模型的通用性相结合——使得构建既经济高效又能力强大的智能体成为可能。

最终，我们观察到，将范式从以大语言模型为中心转向以小语言模型优先的架构，对许多人而言不仅代表了一种技术上的改进，也是一种休谟式的道德义务。随着人工智能界努力应对不断上升的基础设施成本和环境问题，在智能体工作流程中采用并规范小语言模型的使用，可以在促进负责任和可持续的人工智能部署方面发挥关键作用。

3 Position Arguments

我们通过以下非排他性论点来支持视图 **V1–V3**。

3.1 SLMs are already sufficiently powerful for use in agents

- A1** 小型语言模型足够强大，可以在智能体系统中取代大型语言模型。这一论点支持了视图 **V1**。
- A2 Microsoft Phi series.**
- A3 NVIDIA Nemotron-H family.** 2/4.8/90 亿参数的混合 Mamba-Transformer 模型在指令遵循和代码生成准确性上，可与同代密集的 300 亿参数大语言模型相媲美，而推理所需的 FLOPs 仅为其中一个数量级的一小部分 [9]。
- A4 Huggingface SmoLLM2 series.** SmoLLM2 系列紧凑语言模型的参数规模从 1.25 亿到 17 亿不等 [6]，它们在语言理解、工具调用和指令遵循性能上，各自提升至与 140 亿参数的当代模型相当，同时匹敌两年前的 700 亿参数模型。
- A5 NVIDIA Hymba-1.5B.** 这款 Mamba-注意力混合头的小语言模型展示了最佳的指令准确性，其令牌吞吐量比同等规模的 Transformer 模型高出 $3.5\times$ 倍 [23]。
- A6 DeepSeek-R1-Distill series.** DeepSeek-R1-Distill 是一个推理模型系列，规模为 15-80 亿参数，使用 DeepSeek-R1 生成的样本进行训练 [19]。它们展示了强大的常识推理

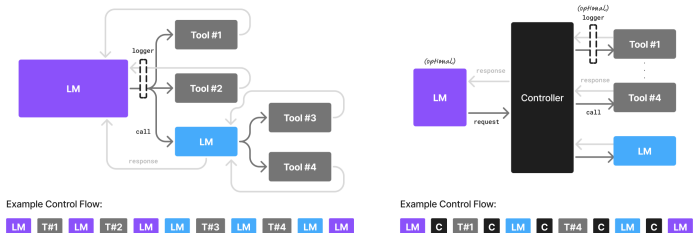


Figure 1: An illustration of agentic systems with different modes of agency. *Left: Language model agency.* The language model acts both as the HCI and the orchestrator of tool calls to carry out a task. *Right: Code agency.* The language model fills the role of the HCI (optionally) while a dedicated controller code orchestrates all interactions.

tool calling, and instruction following performance to 14bn contemporaries while matching 70bn models of 2 years prior.

- **NVIDIA Hymba-1.5B.** This Mamba-attention hybrid-head SLM demonstrates best instruction accuracy and $3.5\times$ greater token throughput than comparably-sized transformer models [23]. On instruction following, it outperforms larger 13bn models.
- **DeepSeek-R1-Distill series.** DeepSeek-R1-Distill is a family of reasoning models featuring 1.5-8bn sizes, trained on samples generated by DeepSeek-R1 [19]. They demonstrate strong commonsense reasoning capabilities. Notably, the DeepSeek-R1-Distill-Qwen-7B model outperforms large proprietary models such as Claude-3.5-Sonnet-1022 and GPT-4o-0513.
- **DeepMind RETRO-7.5B:** Retrieval-Enhanced Transformer (RETRO) is a 7.5bn parameter model augmented with an extensive external text database, achieving performance comparable to GPT-3 (175B) on language modeling while using $25\times$ fewer parameters [10].
- **Salesforce xLAM-2-8B.** The 8bn model achieves state-of-the-art performance on tool calling despite its relatively modest size, surpassing frontier models like GPT-4o and Claude 3.5 [81].

Note that on top of competitive off-the-shelf performance, the reasoning capabilities of SLMs can be enhanced by light-weight selective finetuning [45] or at inference time with self-consistency, verifier feedback, or tool augmentation — e.g., Toolformer (6.7bn) outperforms GPT-3 (175bn) via API use [65], and 1-3bn models have rivaled 30bn+ LLMs on math problems via structured reasoning [84].

In sum, with modern training, prompting, and agentic augmentation techniques, capability — not the parameter count — is the binding constraint. SLMs now supply sufficient reasoning power for a substantial portion of agentic invocations, making them not just viable, but comparatively more suitable than LLMs for modular and scalable agentic systems.

3.2 SLMs are more economical in agentic systems

A2 SLMs are more economical than LLMs in agentic systems. This argument supports view **V3**.

Small models provide significant benefits in cost-efficiency, adaptability, and deployment flexibility. These advantages are specifically valuable in agentic workflows where specialization and iterative refinement are critical. Section 3.1 detailed a number of efficiency comparisons of the listed SLMs to relevant LLMs. Here we draw a more encompassing picture to support argument **A2**.

- **Inference efficiency.** Serving a 7bn SLM is $10\text{--}30\times$ cheaper (in latency, energy consumption, and FLOPs) than a 70–175bn LLM, enabling real-time agentic responses at scale [70, 68, 36, 53]. Recent advances in inference operating systems such as NVIDIA Dynamo [24] explicitly provide support for high-throughput, low-latency SLM inference in

能力。值得注意的是，DeepSeek-R1-Distill-Qwen-7B 模型超越了大型专有模型，如 Claude-3.5-Sonnet-1022 和 GPT-4o-0513。

A7 DeepMind RETRO-7.5B: 检索增强型 Transformer (RETRO) 是一个 75 亿参数的模型，通过一个广泛的外部文本数据库进行增强，在语言建模上实现了与 GPT-3 (1750 亿) 相当的性能，而使用的参数却少了 25 倍 [10]。

A8 Salesforce xLAM-2-8B. 尽管规模相对适中，这款 80 亿参数模型在工具调用方面实现了最先进的性能，超越了 GPT-4o 和 Claude 3.5 等前沿模型 [81]。

A9 在智能体系统中，SLM 比 LLM 更具经济性。此论点支持 视图 **V3**。

A10 Inference efficiency. 部署一个 7bn 参数的 SLM 比部署一个 70–175bn 参数的 LLM 便宜 $10\text{--}30\times$ （在延迟、能耗和 FLOPs 方面），从而能够大规模实现实时智能体响应 [70, 68, 36, 53]。最近的推理操作系统进展，如 NVIDIA Dynamo [24]，明确为云和边缘部署中的高吞吐量、低延迟 SLM 推理提供了支持。此外，由于 SLM 需要较少或无需跨 GPU 和节点的并行化，服务基础设施的维护和运营成本也更低（参见 反驳论点 **A28** 和 论点 **A30**）。

A11 Fine-tuning agility. 对 SLM 进行参数高效（例如 LoRA [33] 和 DoRA [43]）、低资源 [7] 以及全参数微调仅需几个 GPU 小时，使得行为的添加、修复或专业化可以在一夜之间完成，而非数周 [70]。

A12 Edge deployment. 设备端推理系统的进展，如 ChatRTX [59]，展示了在消费级 GPU 上本地执行 SLM 的能力，实现了延迟更低、数据控制更强的实时离线智能体推理。

A13 Parameter and embedding space utilization. 起初，LLM 看似是涉及大量参数的庞然大物，这些参数代表了其输出中大量压缩的信息。然而，仔细观察会发现，流经这些系统的许多嵌入向量非常稀疏，对于任何单个输入仅激活其一小部分参数 [69, 44]，并且实际上是可压缩的 [8, 29]。这种行为在 SLM 中似乎更为收敛 [69, 75]，这表明 SLM 可能本质上更高效，因为其较小比例的参数贡献了推理成本，而对输出没有实质性影响。

A14 与 LLMs 相比，SLMs 具有更强的操作灵活性。此论点支持 视图 **V2** 和 **V3**。

A15 智能体应用是 LM 有限能力子集的接口。这支持了 视图 **V1** 和 **V2**。

A16 智能体交互需要紧密的行为对齐。这与 视图 **V2** 一致。

A17 智能体系统自然地允许其使用的模型选择具有异质性。这与 视图 **V2** 一致。

A18 智能体交互是未来模型改进的良好数据来源。这从根本上支持了 视图 **V2**。

A19 令 $\mathcal{T}$ 为一个使用通用语言的单一任务，并令 L, S 分别为同代的大型和小型语言模型。 L 在 $\mathcal{T}$ 上的性能将始终优于 S 。

A20 有相当多的经验证据表明，在同代模型中，大型语言模型在通用语言理解方面优于小型语言模型。LLM 根据缩放定律 [18] 获得其语言理解能力。其更大的规模使它们能够在广泛的专门自然语言任务中展现出更好的性能，包括文本生成、翻译和推理，超越了以 (a) 相同通用方式训练和 (b) 专门为这些任务从头训练的小模型 [58]。

A21 此外，最近的研究还声称 LLM 拥有一种“语义枢纽”机制，据推测这使它们能够以泛化的方式整合和抽象来自不同模态和语言的语义信息 [80]。如果属实，可以认为 LLM 在跨语言和领域泛化知识方面远比小模型更有效，而同一研究表明小模型缺乏存在此类枢纽的能力 [80]。

A22 流行的缩放定律研究假设模型架构在同一代内保持不变 [32, 31]，而最近关于小语言模型训练的工作表明，针对不同模型规模考虑不同架构具有明显的性能优势 [23, 9]。

both cloud and edge deployments. In addition, since SLMs require less or no parallelization across GPUs and nodes, the maintenance and operation of the serving infrastructure comes at a lower expense as well (see counter-argument **CA4** and argument **A13**).

- **Fine-tuning agility.** Parameter-efficient (e.g., LoRA [33] and DoRA [43]), low-resource [7], and full-parameter finetuning for SLMs require only a few GPU-hours, allowing behaviors to be added, fixed, or specialized overnight rather than over weeks [70].
- **Edge deployment.** Advances in on-device inference systems such as ChatRTX [59] demonstrate local execution of SLMs on consumer-grade GPUs, showcasing real-time, offline agentic inference with lower latency and stronger data control.
- **Parameter and embedding space utilization.** At the outset, LLMs appear to operate as monoliths involving a large amount of parameters representing swathes of compressed information in the production of their outputs. On a closer look, however, many of the embeddings passing through these systems are very sparse, engaging only a fraction of their parameters for any single input [69, 44] and being effectively compressible [8, 29]. That this behavior appears to be more subdued in SLMs [69, 75] suggests that SLMs may be fundamentally more efficient by the virtue of having a smaller proportion of their parameters contribute to the inference cost without a tangible effect on the output.

Modular system design. The position outlined in [56] presents a thorough argument in favor of composite agentic systems. Here we note that the approach of leveraging several models of varying sizes aligns well with the real-world heterogeneity of agentic tasks and is already slowly being incorporated into major software development frameworks [28]. Furthermore, this newly discovered sense for modularity in the context of agents allows for the easy addition of new skills and the ability to adapt to changing requirements, and is consistent with the push for modularity in language model design [27, 12, 40].

The above-mentioned “Lego-like” composition of agentic intelligence—scaling out by adding small, specialized experts instead of scaling up monolithic models—yields systems that are cheaper, faster to debug, easier to deploy, and better aligned with the operational diversity of real-world agents. When combined with tool calling, caching, and fine-grained routing, SLM-first architectures appear to offer the best path forward for cost-effective, modular, and sustainable agentic AI.

3.3 SLMs are more flexible

- A3** SLMs possess greater operational flexibility in comparison to LLMs. This argument stands in support of views **V2** and **V3**.

Due to their small size and the associated reduction in pre-training and fine-tuning costs (Section 3.2), SLMs are inherently more flexible than their large counterparts when appearing in agentic systems. As such, it becomes much more affordable and practical to train, adapt, and deploy multiple specialized expert models for different agentic routines. This efficiency enables rapid iteration and adaptation, making it feasible to address evolving user needs, including supporting new behaviors, meeting new output formatting requirements, and complying with changing local regulation in selected markets [73, 41, 72].

Democratization. One particularly notable and desirable consequence of SLM flexibility when put in place of LLMs is the ensuing democratization of agents. When more individuals and organizations can participate in developing language models with the aim for deployment in agentic systems, the aggregate population of agents is more likely to represent a more diverse range of perspectives and societal needs. This diversity can then help with reducing the risk of systemic biases and encourage competition and innovation. With more actors entering the field to create and refine models, the field will advance more rapidly [38].

3.4 Agents expose only very narrow LM functionality

- A4** Agentic applications are interfaces to a limited subset of LM capabilities. This supports views **V1** and **V2**.

- A23** 小语言模型（Section 3.2）的灵活性提供了解决方案。一个小语言模型可以轻松针对替代视图 **A19** 的任务 **7** 进行微调，以达到所需的可靠性水平。这在缩放定律研究中未被考虑。

- A24** 推理（或更一般地说，测试时计算缩放；参见 Section 3.2）的成本要低得多。一个小语言模型，在保留其更强的跨设备敏捷性优势的同时，可以合理地预期在推理时能够缩放到所需的可靠性水平。

- A25** 所谓的“语义枢纽”的效用，在语言模型需要处理的任务或输入很复杂时才显现出来。然而，高级的智能体系统要么是整体设计的，要么至少被主动提示以执行复杂问题和输入的分解 [52, 17]。因此，我们反而认为，在智能体系统内调用小语言模型时，任务会被适当地分解为非常简单的子任务，以至于枢纽所带来的任何一般性抽象理解都几乎没有用处。

- A26** 在智能体应用中，专用小型语言模型因其规模小带来的单次推理成本优势，被规模经济效益所超越。

A27

A28

- A29** 近期在推理调度和大型推理系统模块化方面的改进，为单体计算集群提供了前所未有的推理系统灵活性 [85, 60, 50]，这反驳了 反驳论点 **A27** 中表达的传统立场。

- A30** 关于推理基础设施搭建成本的最新分析表明，由于底层技术原因，该成本呈现持续下降趋势 [82, 4]。

- A31** 利用 SLM 的能动世界与利用 LLM 的能动世界都是同样可能的世界，但“LLM 能动世界”在部署实践和优化方面拥有相当大的先发优势，且行业惯性已将创新努力完全导向该方向。

- A32 Large amounts of upfront investment in centralized LLM inference infrastructure.** 如 Section 1 中详述，业界已投入巨额资本押注于集中式大语言模型推理，认为这将是未来提供人工智能服务的主导范式。因此，行业在为此目标构建工具和基础设施方面进展迅速，却忽略了在不久的将来，更分散化的小语言模型或设备端推理可能同样可行这一可能性。

- A33 Use of generalist benchmarks in SLM training, design, and evaluation.** 必须指出，许多关于小语言模型设计和开发的工作都遵循大语言模型的设计轨迹，在开发过程中聚焦于相同的通用基准 [47, 61]。关于这一点，[23] 指出，如果只关注衡量智能体代理效用的基准，所研究的小语言模型很容易超越更大的模型。

- A34 Lack of popular awareness.** 尽管在许多工业场景中更具适用性，小语言模型通常未能获得与大语言模型同等水平的营销强度和媒体关注。

- A35 Secure usage data collection.** 初始步骤涉及部署检测工具来记录所有非人机交互代理调用，捕获输入提示、输出响应、单个工具调用的内容，以及可选的延迟指标以供后续针对性优化。在实施方面，建议建立具有基于角色的访问控制的加密日志管道 [55]，并在存储前对所有数据进行来源匿名化处理 [74]。参见 **logger** 中的 **Figure 1** 以获取图示。

- A36 Data curation and filtering.** 开始通过 步骤 **A35** 的管道收集数据。一旦收集到足够的数据（根据经验法则，1 万到 10 万个示例足以用于小型模型的微调 [5, 22]），就有必要移除任何个人身份信息、受保护健康信息或任何其他特定于应用程序的敏感数据，这些数据一旦用于训练 SLM 专家模型，可能导致跨用户账户的数据泄露。许

An AI agent is essentially a heavily instructed and externally choreographed gateway to a language model featuring a human-computer interface and a selection of tools that, when engaged correctly, do something of utility [73]. From this perspective, the underlying large language model that was engineered to be a powerful generalist is through a set of tediously written prompts and meticulously orchestrated context management restricted to operate within a small section of its otherwise large pallet of skills. Thus, we argue that a SLM appropriately fine-tuned for the selected prompts would suffice while having the above-mentioned benefits of increased efficiency and greater flexibility.

It could be argued back that the careful interfacing with a generalist LLM is necessary for strong performance on the narrow task because of the LLM's better understanding of the broader language and the world (alternative view **AV1**). This is addressed in Section 4.1.

3.5 Agentic interactions necessitate close behavioral alignment

A5 Agentic interactions necessitate close behavioral alignment. This aligns with view **V2**.

A typical AI agent has frequent interactions with code, be it through LM tool calling or by returning output that is to be parsed by a piece of agentic code that makes the LM call [52]. It is essential for the success of these interactions that the generated tool call and the generated output conform to strict formatting requirements imposed on it by the order, typing, and nature of the tool's parameters, and the expectation of the code invoking the LM, respectively. In such cases, it becomes unnecessary for the model to handle multiple different formats (e.g. JSON/XML/Python for tool calls and XML/YAML/Markdown/Latex for output [54]), as only one would be chosen for consistency across the agentic application. It is also undesirable for the model to make the occasional hallucinatory mistake and respond in a format different from that being expected by the "code parts" of the agentic system. It is because of this that the SLM trained with a single formatting decision enforced during its post-training or encouraged through additional fine-tuning at a low cost is preferable over a general-purpose LLM in the context of AI agents.

3.6 Agentic systems are naturally heterogeneous

A6 Agentic systems naturally allow for heterogeneity in the selection of models that they use. This aligns with view **V2**.

A language model can itself be a tool called by another language model. Likewise, every time the agent's code invokes a language model, it can, in principle, choose any language model. This is illustrated in Figure 1. We argue that incorporating multiple language models of different sizes and capabilities for queries or operations of different levels of complexity offers a natural way for the introduction of SLMs. In the context of Figure 1-*Left*, an LLM can be used for the model with the root agency, while a SLM could be used for the subordinate LM. In Figure 1-*Right*, all LLMs could in principle be specialized SLMs: one for conversationality, another one for carrying out controller-defined language modeling tasks.

3.7 Agentic interactions are natural pathways for gathering data for future improvement

A7 Agentic interactions are a good source for data for future model improvement. This is fundamentally supportive of view **V2**.

As noted in Section 3.4, invocations of tools and language models during an agentic process are often accompanied by careful prompting that focuses the language model on delivering the narrow functionality that is required at the time. Each one of these invocations is itself a natural source of data for future improvement (under the necessary assumption that no non-retainable confidential data is being processed). A listener decorating the tool/model call interface can gather specialized instruction data that can later be used to produce a fine-tune an expert SLM and lower the cost of that call in the future (see *logger* in Figure 1). We argue that this avenue is enabled by the architecture of agents [52] and produces high-quality organic data (that can be further post-filtered by considering the overall success of the workflow), thus making the production of expert SLMs to stand in place of LLMs a natural step in agent deployment — not just an auxiliary effort.

多典型的敏感数据类型可以使用流行的数据集准备自动化工具来检测、屏蔽或移除 [64, 62]。特定于应用程序的输入（例如法律或内部文件）通常可以自动进行释义，以混淆命名实体和数字细节，同时不损害总体信息内容 [11, 79, 76]。

A37 Task clustering. 对收集到的提示和代理操作应用无监督聚类技术，以识别反复出现的请求模式或内部代理操作模式 [35, 42, 21]。这些聚类有助于定义 SLM 专业化的候选任务。任务的粒度取决于操作的多样性；常见的例子包括意图识别、数据提取、特定文档类型的摘要，或针对代理可用工具的代码生成。

A38 SLM selection. 对于每个识别出的任务，选择一个或多个候选 SLM。选择标准包括 SLM 的固有能力（例如，指令遵循、推理、上下文窗口大小）、其在相关任务类型基准测试中的表现、其许可协议以及其部署占用空间（内存、计算要求）。Section 3.2 的模型是良好的初始候选。

A39 Specialized SLM fine-tuning. 对于每个选定的任务和相应的 SLM 候选，从步骤 **A36** and **A37** 中收集的经过整理的数据中准备一个特定于任务的数据集。然后，在这些专业化的数据集上微调选定的 SLM。可以利用 PEFT 技术，如 LoRA [34] 或 QLoRA [20]，来降低与微调相关的计算成本和内存需求，使该过程更易于进行。如果资源允许且需要最大程度的适应，也可以考虑完全微调。在某些情况下，使用知识蒸馏可能是有益的，即训练专家 SLM 在特定任务数据集上模仿更强大的通用 LLM 的输出。这有助于将 LLM 的一些更细微的能力迁移到 SLM 上。

A40 Iteration and refinement. 可以定期使用新数据重新训练 SLM 和路由模型，以保持性能并适应不断变化的使用模式。这形成了一个持续改进的循环，根据需要返回到步骤 **A36** 或步骤 **A38**。

过去几年，小型语言模型的能力取得了显著进步。尽管语言模型的缩放定律仍然成立，但模型规模与能力之间的缩放曲线正变得越来越陡峭，这意味着新型小型语言模型的能力已非常接近先前的大型语言模型。事实上，最近的进展表明，精心设计的小型语言模型可以达到甚至超过以往只有大得多的模型才能实现的任务性能。

以下引用的各项研究已与大型模型进行了广泛的比较，但并非所有基准测试评估的能力对于其在智能体环境中的部署都是必需的。在此，我们重点强调它们在常识推理（基本理解能力的指标）、工具调用和代码生成（这两者都是模型 工具/代码接口正确通信能力的指标；参见 Figure 1: [77, 78]）以及指令遵循（正确通过代码 模型接口进行响应的能力；[83]）方面的表现。在每种情况下，如果作者提及，我们也引用其效率提升的数据。

Phi-2（27 亿参数）在常识推理和代码生成得分上与 300 亿参数模型相当，同时运行速度 ~ 快 15× 倍 [37]。Phi-3 small（70 亿）[3] 在语言理解和常识推理方面达到同等水平，其代码生成得分可媲美同代高达 700 亿参数的模型。在指令遵循方面，它超越了更大的 130 亿参数模型。

值得注意的是，除了具有竞争力的开箱即用性能外，小语言模型的推理能力可以通过轻量级选择性微调 [45] 或在推理时通过自洽性、验证器反馈或工具增强来提升——例如，Toolformer（67 亿）通过 API 使用超越了 GPT-3（1750 亿）[65]，而 1-30 亿参数的模型通过结构化推理在数学问题上已能与 300 亿+ 的大语言模型相匹敌 [84]。

总之，借助现代的训练、提示和智能体增强技术，能力——而非参数数量——才是制约因素。小语言模型现在为相当一部分智能体调用提供了足够的推理能力，使得它们不仅在模块化和可扩展的智能体系统中可行，而且相比大语言模型更具优势。

4 Alternative Views

The following significant alternative views have been expressed in the academic and popular literature.

4.1 LLM generalists will always have the advantage of more general language understanding

AV1 Let $\mathcal{T}$ be a single task using general language and let L, S be a large and a small language model of the same generation, respectively. The performance of L on $\mathcal{T}$ will always trump that of S .

This alternative view disputes view **V2** and rests on the following counter-arguments:

CA1 There is a non-negligible body of empirical evidence of the superiority of large language models in general language understanding over small language models of the same generation. LLMs acquire their language understanding capabilities in accordance with scaling laws [18]. Their larger scale then enables them to demonstrate better performance across a wide array of specialized natural language tasks, including text generation, translation, and reasoning, outperforming small models trained both (a) in the same general fashion and (b) from scratch specifically for these tasks [58]. It can then be said that to claim otherwise is to contradict the LM scaling laws [32, 31].

CA2 Moreover, recent studies also purport that LLMs possess a “semantic hub” mechanism, which has been hypothesized to enable them to integrate and abstract semantic information from various modalities and languages in a generalized manner [80]. If true, LLMs could be thought to generalize knowledge across languages and domains far more effectively than smaller models, which under the same study lack the capacity for the presence of such a hub [80]. It can then be argued that while small language models may be efficient for narrowly defined or highly specialized tasks, their limited scale fundamentally restricts their ability to achieve the same level of general language understanding in these specialized areas as LLMs because of the lack of room for the internalization of complex abstractions.

A conclusion can be then drawn that LLM generalist models will always retain the advantage of universally better performance on language tasks, no matter how narrowly defined, over small language models of the same generation. This to be to their advantage over SLMs when deployed in agentic applications.

Rebuttal. The above alternative view is the most popularly cited belief against the use of SLMs, even when only a narrow language task needs to be performed [2, 71, 30, 1].

We believe that counter-argument **CA1** is too limited to attack view **V2**, namely because

- A8** Popular scaling law studies assume the model architecture to be kept constant [32, 31] within the same generation, whereas the recent work on small language model training demonstrates that there are distinct performance benefits to considering different architectures for different model sizes [23, 9].
- A9** The flexibility of small language models (Section 3.3) comes to the rescue. A small language model can be easily fine-tuned for the task $\mathcal{T}$ of alternative view **AV1** to perform to the desired level of reliability. This is unaccounted for in scaling law studies.
- A10** Reasoning (or, more generally, test-time compute scaling; see Section 3.2) is significantly more affordable. A small language model, still retaining its benefits of greater cross-device agility can be reasonably expected to be scalable at inference time to the desired level of reliability.

We also believe that counter-argument **CA2** is too arcane to attack view **V2** because

- A11** The utility of the purported “semantic hub” shows itself when tasks or inputs at hand to be processed by the LM are complex. However, advanced agentic systems are either designed in their entirety or at least actively prompted to perform decompositions of complex problems and inputs [52, 17]. Therefore, we argue to the contrary that invocations of small language models within agentic systems would be on appropriately broken-down into sub-tasks so simple that any general abstract understanding due to the hub would be of little utility.

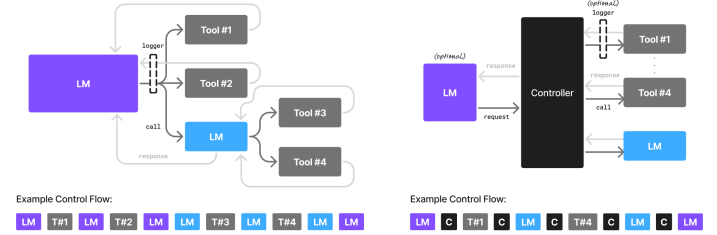


图 1: 不同代理模式的智能体系统示意图。Left: *Language model agency*. 语言模型同时充当人机交互界面和工具调用的编排器来执行任务。Right: *Code agency*. 语言模型（可选地）承担人机交互界面的角色，而一个专用的控制器代码编排所有交互。

3.2 SLMs are more economical in agentic systems

小型模型在成本效益、适应性和部署灵活性方面提供了显著优势。这些优势在专业化和迭代优化至关重要的智能体工作流中尤其有价值。Section 3.1 详细列出了所列 SLM 与相关 LLM 的若干效率比较。在此，我们描绘一个更全面的图景以支持论点 **A9**。

Modular system design. [56] 中概述的立场为复合智能体系统提出了一个全面的论证。我们在此指出，利用多种不同规模模型的方法与智能体任务在现实世界中的异质性非常契合，并且已经逐步被纳入主要的软件开发框架 [28]。此外，这种在智能体背景下新发现的模块化意识，允许轻松添加新技能并适应不断变化的需求，这与语言模型设计中推动模块化的趋势是一致的 [27, 12, 40]。

上述这种“乐高式”的智能体智能组合——通过添加小型、专门的专家来横向扩展，而非纵向扩展单一的大型模型——所产生的系统成本更低、调试更快、部署更容易，并且与现实世界智能体操作的多样性更加契合。当与工具调用、缓存和细粒度路由相结合时，SLM 优先的架构似乎为构建成本效益高、模块化且可持续的智能体 AI 提供了最佳的前进路径。

由于其规模小以及随之而来的预训练和微调成本降低（Section 3.2），当出现在智能体系统中时，SLMs 本质上比其大型对应物更具灵活性。因此，为不同的智能体流程训练、适配和部署多个专门的专家模型变得更加经济实惠和实用。这种效率使得快速迭代和适应成为可能，从而能够满足不断变化的用户需求，包括支持新行为、满足新的输出格式要求以及遵守选定市场中不断变化的本地法规 [73, 41, 72]。

Democratization. 当用 SLMs 替代 LLMs 时，其灵活性带来的一个特别显著且令人期待的后果是随之而来的智能体民主化。当更多的个人和组织能够参与开发旨在部署于智能体系统的语言模型时，智能体的总体群体更有可能代表更多样化的视角和社会需求。这种多样性有助于降低系统性偏见风险，并鼓励竞争与创新。随着更多参与者进入该领域来创建和完善模型，该领域将发展得更快 [38]。

3.3 Agents expose only very narrow LM functionality

AI 智能体本质上是一个经过大量指令指导和外部编排的、通往语言模型的网关，它具有人机界面和一系列工具，当正确使用时，能产生实用价值 [73]。从这个角度来看，底层的大型

4.2 LLM inference will still be cheaper because of their centralization

AV2 The per-token inference cost benefit of the smallness of specialized SLMs in agentic applications is dwarfed by the economy of scale.

It could be argued that the analysis in argument **A2** that put forth in favor of view **V3** was ignorant of the wider business of AI model deployment:

CA3 It is more difficult to fully utilize and properly balance the load for an expert SLM inference endpoint than it is for a generalist LLM endpoint [70, 25].

CA4 The costs of inference infrastructure setup combined with the costs of acquiring and retaining talent for its upkeep are often omitted in inference cost calculations but figure more prominently if the deployment of (S)LMs became the responsibility of the agent service developer. Early industrial reports point to considerable costs associated with these operations [39, 13, 67].

Acknowledgment. We acknowledge that alternative view **AV2** is a valid view, with the exact economical considerations being highly case-specific. We believe that the jury is still out on alternative view **AV2**, but that several factors hint that view **V3** might prevail:

A12 Recent improvements in inference scheduling and large inference system modularization offer unprecedented levels of inference system flexibility in monolithic computing clusters [85, 60, 50], countering the traditional stance expressed in counter-argument **CA3**.

A13 The most recent analyses on the set-up costs of inference infrastructure indicate a consistent falling trend due to underlying technological reasons [82, 4].

4.3 Equally possible worlds

AV3 Both the agentic world utilizing SLMs and agentic world utilizing LLMs are equally possible worlds, but the “LLM agentic world” has a considerable head start in terms of deployment practice and optimization, and the industry inertia already funnels efforts into innovation solely in that direction.

Acknowledgment. We acknowledge alternative view **AV3** as a distinct possibility, but maintain the position that the weight of advantages described across arguments **A1–A7** can plausibly overturn the present state of affairs.

5 Barriers to Adoption

It would be prudent to ask oneself: If the arguments **A1–A7** are truly compelling, why do the ever newer generations of agents seemingly just perpetuate the status quo of using generalist LLMs?

We believe the following to be among the today’s main barriers to wide-spread adoption of SLMs:

B1 Large amounts of upfront investment in centralized LLM inference infrastructure. As detailed in Section 1, large capital bets have been made on the centralized LLM inference being the leading paradigm in providing AI services in the future. As such, the industry has been much quicker at building the tools and infrastructure to that end, omitting any considerations for the possibility that more decentralized SLM or on-device inference might be equally feasible in the near future.

B2 Use of generalist benchmarks in SLM training, design, and evaluation. It must be pointed out that much of the work on SLM design and development follows the tracks of LLM design, focusing on the same generalist benchmarks in their development [47, 61]. On this point, [23] notes that if one focuses solely on benchmarks measuring the agentic utility of agents, the studied SLMs easily outperform larger models.

B3 Lack of popular awareness. SLMs often do not receive the level of marketing intensity and press attention LLMs do, despite their better suitability in many industrial scenarios.

语言模型被设计成一个强大的通才，但通过一系列精心编写的提示词和细致编排的上下文管理，其操作被限制在其原本庞大的技能库中的一小部分范围内。因此，我们认为，一个针对所选提示词进行适当微调的 SLM 就足够了，同时还能获得上述提到的提高效率和增强灵活性的好处。

可能会有人反驳说，与通才 LLM 的精心接口对于在狭窄任务上取得强劲性能是必要的，因为 LLM 对更广泛的语言和世界有更好的理解（替代视图 **A19**）。这一点在 Section 4.1 中进行了讨论。

3.4 Agentic interactions necessitate close behavioral alignment

一个典型的 AI 智能体频繁地与代码交互，无论是通过 LM 工具调用，还是通过返回需要由调用 LM 的智能体代码解析的输出 [52]。这些交互成功的关键在于，生成的工具调用和生成的输出必须严格遵守其格式要求，这些要求分别由工具参数的顺序、类型和性质，以及调用 LM 的代码的期望所决定。在这种情况下，模型处理多种不同格式（例如，工具调用的 JSON/XML/Python 和输出的 XML/YAML/Markdown/Latex [54]）变得不必要，因为为了智能体应用的一致性，只会选择一种格式。同样，模型偶尔产生幻觉性错误并以不同于智能体系统“代码部分”所期望的格式进行响应，也是不可取的。正因如此，在 AI 智能体的背景下，通过后训练强制执行单一格式决策，或以低成本通过额外微调鼓励该决策的 SLM，优于通用目的的 LLM。

3.5 Agentic systems are naturally heterogeneous

一个语言模型本身可以是另一个语言模型调用的工具。同样，每次智能体的代码调用语言模型时，原则上它可以选择任何语言模型。这在 Figure 1 中有所说明。我们认为，针对不同复杂程度的查询或操作，纳入不同规模和能力的多个语言模型，为引入 SLM 提供了一种自然的方式。在 Figure 1-*Left* 的背景下，LLM 可用于具有根代理权的模型，而 SLM 可用于从属的 LM。在 Figure 1-*Right* 的背景下，原则上所有 LM 都可以是专门的 SLM：一个用于对话性，另一个用于执行控制器定义的语言建模任务。

3.6 Agentic interactions are natural pathways for gathering data for future improvement

如 Section 3.3 所述，在智能体过程中调用工具和语言模型时，通常伴随着精心设计的提示，这些提示旨在将语言模型聚焦于交付当时所需的特定功能。每一次这样的调用本身都是未来改进的自然数据来源（基于必要的假设，即不处理不可保留的机密数据）。一个装饰在工具/模型调用接口上的监听器可以收集专门的指令数据，这些数据随后可用于微调专家小型语言模型，并在未来降低该调用的成本（参见 Figure 1 中的 **logger**）。我们认为，这种途径由智能体的架构 [52] 所实现，并产生高质量的有机数据（可以通过考虑工作流程的整体成功进行进一步的后过滤），从而使生产专家小型语言模型来替代大型语言模型成为智能体部署的一个自然步骤——而不仅仅是一项辅助工作。

4 Alternative Views

学术和大众文献中表达了以下重要的不同观点。

4.1 LLM generalists will always have the advantage of more general language understanding

此替代观点对 视图 **V2** 提出质疑，并基于以下反驳论点：

We note that barriers **B1–B3** are practical hurdles and far from being fundamental flaws of the SLM technology in the context of agentic AI. With advanced inference scheduling systems such as Dynamo [24], barrier **B1** is being reduced to a mere effect of inertia. barrier **B2** is becoming increasingly recognized in the field [23, 37], and it would be natural for barrier **B3** to fall once the economic benefits of SLM deployment in agentic applications (argument **A2**) are better known. With the inertia of barrier **B1** in particular, we do not endeavor to give a timeline for the retreat of these barriers or the popular adoption of SLMs.

6 LLM-to-SLM Agent Conversion Algorithm

The very nature of agentic applications enables them to eventually switch from using LLM generalists to using SLM specialists at many of their interfaces. In the following steps, we outline an algorithm that describes one possible way to carry out the change of the underlying model painlessly.

- S1 Secure usage data collection.** The initial step involves deploying instrumentation to log all non-HCI agent calls, capturing input prompts, output responses, contents of individual tool calls, and optionally latency metrics for a later targeted optimization. In terms of implementation, it is the recommended practice to set up encrypted logging pipelines with role-based access controls [55] and anonymize all data with respect to its origins before storage [74]. See `logger` in Figure 1 for an illustration.
- S2 Data curation and filtering.** Begin collecting data through the pipelines of step **S1**. Once a satisfactory amount of data has been collected (10k-100k examples being sufficient for fine-tuning of small models as a rule of thumb [5, 22]), it is necessary to remove any PII, PHI, or any other application-specific sensitive data that could cause a data leak across user accounts once used to produce a SLM specialist. Many typical varieties of sensitive data can be detected and masked or removed using popular automated tools for dataset preparation [64, 62]. Application specific inputs (e.g. legal or internal documents) can be often be automatically paraphrased to obfuscate named entities and numerical details without compromising the general information content [11, 79, 76].
- S3 Task clustering.** Employ unsupervised clustering techniques on the collected prompts and agent actions to identify recurring patterns of requests or internal agent operations [35, 42, 21]. These clusters help define candidate tasks for SLM specialization. The granularity of tasks will depend on the diversity of operations; common examples include intent recognition, data extraction, summarization of specific document types, or code generation with respect to tools available to the agent.
- S4 SLM selection.** For each identified task, select one or more candidate SLMs. Criteria for selection include the SLM’s inherent capabilities (e.g., instruction following, reasoning, context window size), its performance on relevant benchmarks for the task type, its licensing, and its deployment footprint (memory, computational requirements). Models of Section 3.2 serve as good starting candidates.
- S5 Specialized SLM fine-tuning.** For each selected task and corresponding SLM candidate, prepare a task-specific dataset from the curated data collected in steps **S2** and **S3**. Then, fine-tune the chosen SLMs on these specialized datasets. PEFT techniques such as LoRA [34] or QLoRA [20] can be leveraged to reduce computational costs and memory requirements associated with fine-tuning, making the process more accessible. Full fine-tuning can also be considered if resources permit and maximal adaptation is required. In some cases, it may be beneficial to use knowledge distillation, where the specialist SLM is trained to mimic the outputs of the more powerful generalist LLM on the task-specific dataset. This can help transfer some of the more nuanced capabilities of the LLM to the SLM.
- S6 Iteration and refinement.** One may retrain the SLMs and the router model periodically with new data to maintain performance and adapt to evolving usage patterns. This forms a continuous improvement loop, returning to step **S2** or step **S4** as appropriate.

7 Call for Discussion

The agentic AI industry is showing the signs of a promise to have a transformative effect on white collar work and beyond.

因此可以说，主张相反的观点就是与语言模型缩放定律 [32, 31] 相矛盾。因此可以认为，虽然小型语言模型对于定义狭窄或高度专门化的任务可能很高效，但其有限的规模从根本上限制了它们在实现与 LLM 相同水平的通用语言理解方面的能力，因为它们缺乏内化复杂抽象的空间。

由此可以得出结论：在同代模型中，LLM 通用模型在语言任务上的性能将始终保持普遍优于小型语言模型的优势，无论任务定义多么狭窄。这使得它们在应用于智能体应用时相对于 SLM 具有优势。

Rebuttal. 上述替代观点是反对使用 SLM 时最常被引用的观点，即使只需要执行一个狭窄的语言任务时也是如此 [2, 71, 30, 1]。

我们还认为 反驳论点 **A21** 过于晦涩，不足以攻击 视图 **V2**，因为

4.2 LLM inference will still be cheaper because of their centralization

可以认为，论点 **A9** 中提出的支持 视图 **V3** 的分析忽略了 AI 模型部署的更广泛业务现实：

与通用大语言模型端点相比，专用小型语言模型推理端点的资源充分利用和负载均衡更为困难 [70, 25]。推理基础设施的搭建成本，加上获取和留住维护人才的成本，在推理成本计算中常被忽略；但如果（小型）语言模型的部署成为智能体服务开发者的责任，这些成本将变得更为突出。早期的行业报告指出，这些运营活动伴随着相当大的成本 [39, 13, 67]。

Acknowledgment. 我们承认 替代视图 **A26** 是一个合理的观点，其具体的经济考量高度依赖于具体案例。我们认为关于 替代视图 **A26** 的结论尚未最终确定，但有若干因素暗示 视图 **V3** 可能会占据上风：

4.3 Equally possible worlds

Acknowledgment. 我们承认替代视图 **A31** 是一种明显的可能性，但坚持认为，贯穿论点 **A1**, **A9** and **A14–A18** 所描述的优势权重，足以合理地扭转当前局面。

5 Barriers to Adoption

审慎地扪心自问：如果论点 **A1**, **A9** and **A14–A18** 确实具有说服力，为何不断更新的智能体世代似乎只是在延续使用通用大语言模型的现状？

我们认为，以下是当今阻碍小语言模型广泛采用的主要障碍：

我们注意到，障碍 **A32–A34** 是实践中的障碍，远非小语言模型技术在代理式人工智能背景下的根本缺陷。借助诸如 Dynamo [24] 这样的先进推理调度系统，障碍 **A32** 正在减弱为仅仅是惯性的影响。障碍 **A33** 在该领域正日益得到认可 [23, 37]，一旦小语言模型在代理式应用中部署的经济效益（论点 **A9**）更为人所知，障碍 **A34** 自然也会消失。特别是考虑到障碍 **A32** 的惯性，我们并不试图为这些障碍的消退或小语言模型的普及采用设定一个时间表。

6 LLM-to-SLM Agent Conversion Algorithm

智能体应用的本质使其最终能够在许多接口处，从使用通用大语言模型转向使用专用小语言模型。在接下来的步骤中，我们概述了一种算法，该算法描述了一种实现底层模型无缝切换的可能方式。

It is the view of the authors that any expense savings or improvements on the sustainability of AI infrastructure would act as a catalyst for this transformation, and that it is thus eminently desirable to explore all options for doing so.

We therefore call for both contributions to and critique of our position, to be directed to agents@nvidia.com, and commit to publishing all such correspondence at research.nvidia.com/labs/lpr/slm-agents.

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7 Call for Discussion

智能体人工智能产业正显示出其有望对白领工作及其他领域产生变革性影响的迹象。

作者们认为，任何在人工智能基础设施成本节约或可持续性方面的改进，都将成为这一变革的催化剂，因此，探索实现这一目标的所有方案是极其可取的。

因此，我们呼吁各方对我们的立场提出贡献或批评，请发送至 agents@nvidia.com，并承诺将所有此类通信发表于 research.nvidia.com/labs/lpr/slm-agents。

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A Definitions

This appendix provides two justifications for the choice of definitions in Section 2.1.

A.1 Pragmatic argument

It is desirable to have a definition of SLMs that meets three key criteria:

- **Timelessness.** The definition should be timeless: It should avoid dependence on hardware-specific metrics like parameter count or FLOPs, which quickly become obsolete as technology advances—what qualifies as “small” today may be “large” tomorrow.
- **Practicality.** The definition is likely to have much wider generality if it is grounded in practical use, reflecting the real-world goal of deploying SLMs on widely available consumer devices, where they can serve the user in their proximity with low-latency inference.
- **Motivation alignment.** The definition should capture the fundamental motivation that drives the training of SLMs in the first place, which is to enable capable language models that can run on-device or within significantly constrained budgets compared to LLMs.

We find definition **WD1** to possess all three. Definition **WD2** is then phrased to complement the set of all language models.

A.2 Limit argument

To explore the distinction between small and large language models in the context of agentic AI, let us adopt the uncompromising lens of an extremist, for whom intelligence must be either maximally small or maximally large.

Imagine a super-intelligent system spanning galactic scales, marshaling all available matter to optimize its computations. Such a system, while theoretically capable of addressing profound questions would face insurmountable physical constraints. The speed of light limits communication, with round-trip delays across a galaxy potentially spanning tens of thousands of years [63]. This latency precludes real-time coordination, fragmenting the system into loosely coupled components rather than a unified “mind”. At cosmological scales, spanning millions or billions of light-years, communication delays could approach or exceed the universe’s age of 13.8 billion years [15]. Such a system, while vast, would be impractical for human-relevant applications, its computations unfolding over eons.

Conversely, consider an infinitely small intelligent system, reduced to the minimal substrate capable of computation. Such a system, akin to the simplest biological organisms, would lack the sensors, effectors, or computational capacity to meaningfully interact with its environment. Its intelligence would be constrained to rudimentary evolution, much like early life forms that emerged 3.5 billion years ago [66]. Yet, even in nature, scale varies dramatically: living organisms range from bacteria (hundreds of nanometers) to blue whales (up to 30 meters), the heaviest ones being limited by heat dissipation due to their high volume-to-surface ratio [26]. At the cosmic scale, all terrestrial life appears microscopic, suggesting that absolute size is less critical than functional adaptability.

Hereby: Humans, often regarded as a pinnacle of intelligence, offer a useful anchor for defining SLMs and LLMs. With a brain-to-body mass ratio surpassed only by small mammals like mice [49], humans balance computational efficiency with practical embodiment. SLMs, by analogy, are systems compact enough to run on personal devices, be trained with modest human interaction, or perform constrained, verifiable tasks. LLMs, in contrast, demand datacenter-scale infrastructure, organization-level training, and extensive validation, reflecting their computational load. The extremist perspective hints at a profound truth: intelligence is not defined by size alone but by the balance of capability, efficiency, and context. For agentic workflows, SLMs may offer agility and accessibility, while LLMs provide depth at the cost of scale.

It is because of this apparent continuum that, if pressed to provide a *definition* of SLMs, we choose to anchor it in characteristics of a model that can be deployed in a distributed fashion with present-day technology and be interactive enough when engaging with a human to be of utility. Proceeding in such a way, the contemporary instances of the definition will evolve as the technology underpinning these models advances, making the definition sufficiently timeless to be practical.

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B LLM-to-SLM Replacement Case Studies

This appendix assesses the potential extent of replacing large language model invocations with small language models in three popular open-source agents: *MetaGPT*, *Open Operator*, and *Cradle*. Each case study examines the use of LLMs, evaluates where SLMs may be viable replacements, and concludes with an estimated percentage of replaceable queries.

B.1 Case study 1: MetaGPT

Name. MetaGPT

License. Apache 2.0

Purpose. MetaGPT is a multi-agent framework designed to emulate a software company. It assigns roles such as Product Manager, Architect, Engineer, and QA Engineer to collaboratively handle tasks including requirement drafting, system design, implementation, and testing.

LLM Invocations.

- *Role-Based Actions.* Each agent role invokes LLMs to fulfill its specialized responsibilities (e.g., coding, documentation).
- *Prompt Templates.* Structured prompts used for consistent outputs.
- *Dynamic Intelligence.* Used for planning, reasoning, and adaptation.
- *Retrieval-Augmented Generation (RAG).* Retrieves relevant documents to enhance generation.

Assessment for SLM Replacement. SLMs would be well-suited for routine code generation and boilerplate tasks, as well as for producing structured responses based on predefined templates. However, they would require further fine-tuning data to reliably perform more complex tasks, such as architectural reasoning and adaptive planning or debugging, which would initially benefit from the broader contextual understanding and the generality of LLMs.

Conclusion. In the case of MetaGPT, we estimate that about 60% of its LLM queries could be reliably handled by appropriately specialized SLMs.

B.2 Case study 2: Open Operator

Name. Open Operator

License. MIT License

Purpose. Open Operator is a workflow automation agent enabling users to define behaviours of agents that can perform tasks like API calls, monitoring, and orchestration using tools and services.

LLM Invocations

- *Natural Language Processing.* Parses user intent.
- *Decision Making.* Guides execution flow.
- *Content Generation.* Writes summaries, reports.

Assessment for SLM Replacement SLMs would be well-suited for tasks such as simple command parsing and routing, as well as generating messages based on predefined templates. They could be meeting their limitations when dealing with more complex tasks that would require multi-step reasoning or the ability to maintain conversation flow and context over time—areas where LLMs would continue to offer significant advantages.

A Definitions

本附录为 Section 2.1 中的定义选择提供了两项理由。

A.1 Pragmatic argument

一个理想的小型语言模型定义应满足三个关键标准：

- **Timelessness.** 定义应具有时效性：它应避免依赖于硬件特定的指标，如参数量或浮点运算次数，这些指标会随着技术进步而迅速过时——今天被认为是“小”的，明天可能就变成了“大”的。
- **Practicality.** 如果定义基于实际应用，则可能具有更广泛的普适性，这反映了在广泛可用的消费设备上部署小型语言模型的现实目标，使其能够在用户附近提供低延迟推理服务。
- **Motivation alignment.** 定义应抓住驱动小型语言模型训练的根本动机，即实现能够在设备上运行或在相比大型语言模型显著受限的预算内运行的有能力的语言模型。

我们认为定义 **WD1** 具备所有这三项。定义 **WD2** 的表述则是为了补充所有语言模型的集合。

A.2 Limit argument

为了在智能体人工智能的背景下探索小型与大型语言模型之间的区别，让我们采用极端主义者毫不妥协的视角，对他们而言，智能必须是极致的微小或极致的庞大。

设想一个横跨星系尺度的超级智能系统，它调动所有可用物质来优化其计算。这样的系统，虽然在理论上能够解决深刻的问题，但将面临无法逾越的物理限制。光速限制了通信，跨越一个星系的往返延迟可能长达数万年 [63]。这种延迟排除了实时协调的可能性，将系统分裂成松散耦合的组件，而非一个统一的“心智”。在宇宙学尺度上，跨越数百万或数十亿光年，通信延迟可能接近甚至超过宇宙 138 亿年的年龄 [15]。这样的系统虽然庞大，但对于人类相关的应用来说是不切实际的，其计算将跨越漫长的纪元。

相反，考虑一个无限小的智能系统，缩减到能够进行计算的最小基底。这样的系统，类似于最简单的生物有机体，将缺乏传感器、效应器或有意义地与周围环境互动的计算能力。其智能将被限制在原始的进化水平，就像 35 亿年前出现的早期生命形式一样 [66]。然而，即使在自然界中，尺度也差异巨大：生物有机体从细菌（数百纳米）到蓝鲸（可达 30 米）不等，最重的生物因其高体积表面积比而受到散热限制 [26]。在宇宙尺度上，所有地球生命都显得极其微小，这表明绝对尺寸不如功能适应性关键。

因此：人类，通常被视为智能的顶峰，为定义 SLM 和 LLM 提供了一个有用的锚点。人类的大脑与身体质量比仅被小鼠等小型哺乳动物超越 [49]，他们在计算效率与实际具身化之间取得了平衡。类比而言，SLM 是足够紧凑、可以在个人设备上运行、通过适度的人类交互进行训练或执行受限、可验证任务的系统。相比之下，LLM 则需要数据中心规模的基础设施、组织级别的训练以及广泛的验证，这反映了它们的计算负载。极端主义者的视角暗示了一个深刻的真理：智能并非仅由大小定义，而是由能力、效率和情境的平衡所定义。对于智能体工作流，SLM 可能提供敏捷性和可访问性，而 LLM 则以规模为代价提供深度。

正是由于这种明显的连续性，如果必须提供一个 SLM 的 *definition*，我们选择将其锚定在这样一个模型的特性上：该模型能够利用现有技术以分布式方式部署，并且在与人交互时足够互动以产生效用。以这种方式进行，该定义的当代实例将随着支撑这些模型的技术进步而演变，从而使该定义足够永恒且实用。

Conclusion. In the case of Open Operator, we estimate that about 40% of its LLM queries could be reliably handled by appropriately specialized SLMs.

B.3 Case study 3: Cradle

Name. Cradle

License. MIT License

Purpose Cradle is designed for General Computer Control (GCC), enabling agents to operate GUI applications via screenshot input and simulated user interaction.

LLM Invocations.

- *Interface Interpretation.* Understands visual context.
- *Task Execution Planning.* Determines sequences of GUI actions.
- *Error Handling.* Diagnoses and reacts to unexpected software states.

Assessment for SLM Replacement SLMs would be well-suited for handling repetitive GUI interaction workflows and the execution of pre-learned click sequences. However, they would face challenges when it comes to tasks involving dynamic GUI adaptation or unstructured error resolution, which would require a higher degree of contextual understanding typically provided by LLMs.

Conclusion In the case of Cradle, we estimate that about 70% of its LLM queries could be reliably handled by appropriately specialized SLMs.

B LLM-to-SLM Replacement Case Studies

本附录评估了三个流行的开源智能体（*MetaGPT*、*Open Operator* 和 *Cradle*）中，用小型语言模型替代大型语言模型调用的潜在范围。每个案例研究都考察了 LLM 的使用情况，评估了 SLM 可能成为可行替代方案的地方，并最终给出了可替换查询的估计百分比。

B.1 Case study 1: MetaGPT

Name. MetaGPT

License. Apache 2.0

Purpose. MetaGPT 是一个旨在模拟软件公司的多智能体框架。它分配了诸如产品经理、架构师、工程师和 QA 工程师等角色，以协作处理包括需求起草、系统设计、实现和测试在内的任务。

LLM Invocations.

- *Role-Based Actions.* 每个智能体角色调用 LLM 来完成其专业职责（例如，编码、文档编写）。
- *Prompt Templates.* 使用结构化提示来确保输出的一致性。
- *Dynamic Intelligence.* 用于规划、推理和适应。
- *Retrieval-Augmented Generation (RAG).* 检索相关文档以增强生成效果。

Assessment for SLM Replacement. SLM 非常适合处理常规的代码生成和样板任务，以及基于预定义模板生成结构化响应。然而，它们需要更多的微调数据才能可靠地执行更复杂的任务，例如架构推理、自适应规划或调试，这些任务最初将受益于 LLM 更广泛的上下文理解和通用性。

Conclusion. 在 MetaGPT 的案例中，我们估计其约 60% 的 LLM 查询可以由经过适当专门化训练的 SLM 可靠处理。

B.2 Case study 2: Open Operator

Name. Open Operator

License. MIT License

Purpose. Open Operator 是一个工作流自动化智能体，使用户能够定义智能体的行为，这些智能体可以利用工具和服务执行 API 调用、监控和编排等任务。

LLM Invocations

- *Natural Language Processing.* 解析用户意图。
- *Decision Making.* 指导执行流程。
- *Content Generation.* 撰写摘要、报告。

Assessment for SLM Replacement SLM 非常适合诸如简单的命令解析和路由，以及基于预定义模板生成消息等任务。在处理需要多步推理或需要随时间推移保持对话流和上下文等更复杂的任务时，它们可能会遇到局限——在这些领域，LLM 将继续提供显著优势。

Conclusion. 在 Open Operator 的案例中，我们估计其约 40% 的 LLM 查询可以由经过适当专门化训练的 SLM 可靠处理。

B.3 Case study 3: Cradle

Name. 摇篮

License. MIT 许可证

Purpose 摇篮 (Cradle) 专为通用计算机控制 (GCC) 而设计，使智能体能够通过截图输入和模拟用户交互来操作图形用户界面应用程序。

LLM Invocations.

- *Interface Interpretation.* 理解视觉上下文。
- *Task Execution Planning.* 确定图形用户界面操作序列。
- *Error Handling.* 诊断并对意外软件状态做出反应。

Assessment for SLM Replacement 专用语言模型 (SLMs) 非常适合处理重复性的图形用户界面交互工作流和执行预先习得的点击序列。然而，当涉及需要动态图形用户界面适应或非结构化错误解决的任务时，它们将面临挑战，因为这些任务通常需要由大型语言模型 (LLMs) 提供的更高层次的上下文理解。

Conclusion 在摇篮 (Cradle) 的案例中，我们估计其约 70% 的大型语言模型查询可以由经过适当专门化的专用语言模型可靠地处理。